

Multi-Digit Addition & Mental Strategies

Adding and subtracting large numbers confidently—written methods and mental shortcuts.



Use these notes well and you will be able to:

- Add and subtract multi-digit numbers using column method.
- Use the number line to verify multi-digit calculations.
- Apply at least five mental strategies for fast calculation.
- Choose the most efficient strategy for a given problem.
- Identify when to use a written method vs a mental method.

If you follow the steps, you will succeed.

Key Words

KEY WORD	SIMPLE DEFINITION
Multi-digit	A number with more than one digit — e.g. 47, 382, 1 056.
Column addition	Written method: align digits by place value, add each column right-to-left.
Carrying	When a column sums to 10 or more, carry 1 into the next column.
Borrowing	In subtraction, take 1 from the next column when the digit is too small.
Compensation	Round to a friendly number, calculate, then adjust by the difference.
Partitioning	Split a number into hundreds, tens, and ones before calculating.
Bridging	Jump to the nearest 10 or 100 first, then continue.
Number bond	A pair of numbers that add to a given target (often 10 or 100).
Inverse operation	The opposite operation — subtraction undoes addition and vice versa.
Estimate	An approximate answer used to check reasonableness.

Written Method – Column Addition

Line up digits by PLACE VALUE (units under units, tens under tens). Add from RIGHT to LEFT. If a column total is ≥ 10 , write the units digit and CARRY the tens digit into the next column.

	H	T	U	Notes
	2	6	8	First number
+	1	7	5	Second number
Carry		1	1	Carry from units and tens columns
TOTAL	4	4	3	Answer: 443

Tip: 268 + 175: **Units:** $8+5=13$ write 3, carry 1. **Tens:** $6+7+1(\text{carry})=14$ write 4, carry 1. **Hundreds:** $2+1+1=4$. **Answer:** 443

Written Method – Column Subtraction (Borrowing)

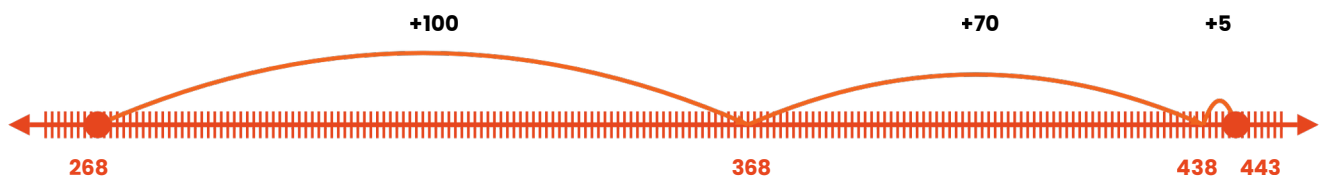
Line up by place value. Subtract RIGHT to LEFT. If the top digit is SMALLER than the bottom digit, BORROW 1 from the column to the left (that column loses 1, the current column gains 10).

	H	T	U	Notes
	5	3	2	Top number (532)
-	1	8	6	Subtract this
Borrow	5	2 ¹²	12	Borrow: T becomes 2, U becomes 12
TOTAL	3	4	6	Answer: 346

Verify Using the Number Line

After a written calculation, use chunked jumps on the number line to CHECK your answer.

Check: $268 + 175$ jump +100, +70, +5



Mental Strategies — Five Essential Methods

Strategy 1: Partitioning (Split into Hundreds, Tens, Units)

Calculate $346 + 253$ by partitioning

Split: $346 = 300 + 40 + 6$ and $253 = 200 + 50 + 3$

Add hundreds: $300 + 200 = 500$

Add tens: $40 + 50 = 90$

Add units: $6 + 3 = 9$

Answer: $500 + 90 + 9 = 599$

Strategy 2: Compensation (Round, Calculate, Adjust)

Calculate $487 + 199$

199 is close to 200. Round up: $487 + 200 = 687$.

We added 1 too many, so subtract 1.

Answer: $687 - 1 = 686$

Calculate $352 - 198$

198 is close to 200. Subtract 200: $352 - 200 = 152$.

We subtracted 2 too many, so add 2 back.

Answer: $152 + 2 = 154$

Strategy 3: Bridging Through 10 or 100

Calculate $76 + 37$ by bridging through 100

76 needs 24 more to reach 100.

Split 37 into $24 + 13$.

$76 + 24 = 100$, then $100 + 13 = 113$.

Answer: $76 + 37 = 113$

Strategy 4: Using Number Bonds to 10 and 100

Bond to 10	Bond to 10	Bond to 100	Bond to 100
$3 + 7 = 10$	$6 + 4 = 10$	$30 + 70 = 100$	$45 + 55 = 100$
$8 + 2 = 10$	$5 + 5 = 10$	$60 + 40 = 100$	$73 + 27 = 100$
$9 + 1 = 10$	$1 + 9 = 10$	$85 + 15 = 100$	$18 + 82 = 100$

Strategy 5: Adding / Subtracting in Parts (Left to Right)

Calculate $574 - 136$ (left to right)

Subtract hundreds first: $574 - 100 = 474$

Subtract tens: $474 - 30 = 444$

Subtract units: $444 - 6 = 438$

Answer: $574 - 136 = 438$

Choosing the Right Strategy

IF the problem looks like...	USE THIS STRATEGY	BECAUSE...
...+ or - a near-round number (99, 198, 301)	Compensation	Rounding makes it easy
...adding numbers close to 10 or 100	Bridging	Clean jumps to landmark
...large numbers with no friendly pattern	Partitioning	Breaks it into steps
...need to check or visualise	Number line	Visual = no mistakes
...any written multi-digit sum	Column method	Systematic and reliable

Practice Questions

COLUMN METHOD

1. Calculate using column addition: $478 + 365$
2. Calculate using column subtraction: $703 - 289$
3. Add: $1294 + 3817$ (show carrying clearly)

MENTAL STRATEGIES

4. Use compensation: $645 + 299$
5. Use compensation: $831 - 399$
6. Use partitioning: $427 + 356$
7. Use bridging: $88 + 45$
8. Use left-to-right: $682 - 248$

MIXED & VERIFICATION

9. Estimate first, then calculate exactly: $523 + 278$
10. A shop had 845 items. 367 were sold. How many remain? Use any strategy.
11. Check using the inverse: If $574 + 268 = 842$, verify by subtracting.

ADD YOUR NOTES HERE



Hey smart student!

There is no single 'best' strategy — the best one is the one that gets YOU to the correct answer fastest and most confidently.

Practise all five methods until you can choose the right one automatically.

Keep going — see you in the next lesson! From FREDi